

# Registral dispersion — recommended parameterization



**Tool:** `Registral_Dispersion` ( `registral-dispersion` v0.3.0)

**Purpose:** advised settings for common research setups in symbolic-score registral dispersion analysis:

**Metric definitions and interpretive limits:** [METRIC SEMANTICS.md](#)

1. **Static vertical aggregate** — the vertical registral “state” at each moment (or over the whole score), without arbitrary temporal smoothing.
2. **Moving fragment** — how registral dispersion **evolves** along the score via sliding temporal windows.
3. **One-number global summary** — duration-weighted whole-score aggregate via summarize CLI/API.

All values below assume **MusicXML / MXL / MIDI** input parsed by **music21**. Results are **symbolic** (not acoustic).

## Shared concepts (both setups)

| Concept                     | Recommended default                                                                                                            | Notes                                                                                                           |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Register preset</b>      | <b>A0–C8 (full notated range)</b>                                                                                              | Results depend on the band.<br>Normalized columns scale by $R = \text{register\_high} - \text{register\_low}$ . |
| <b>Raw vs normalized</b>    | <b>Raw semitones</b> for primary publication;<br><b>normalized_*</b> for cross-score comparison with different register bounds | Normalized values are <b>not</b> perceptual brightness.                                                         |
| <b>Heatmap (complement)</b> | Always include when exploring register                                                                                         | Independent of dispersion formulas; use <b>registral_ember</b> +                                                |

| Concept        | Recommended default                                 | Notes                                                                                                                                                                 |
|----------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <b>notational occupancy</b><br>(informal “density”) | <b>log1p_counts</b> . See <a href="#">METRIC SEMANTICS.md</a> §7.                                                                                                     |
| <b>Exports</b> | Always keep <b>CSV + JSON</b>                       | JSON records <code>analysis_profile</code> ,<br><code>pitch_sampling_mode</code> ,<br><code>observation_mode</code> , formulas, and<br><code>package_version</code> . |

## Analysis profile (choose one stance)

| Profile                                             | Implied sampling                  | Use when                                                                                                                                                                                                               |
|-----------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>occupied_space</b><br>( <i>advised default</i> ) | <code>unique_pitch_heights</code> | You care about <b>geometry of occupied pitch heights</b> (opening/compression of registral space). Doublings at the same pitch do <b>not</b> inflate dispersion.<br><b>Primary read:</b> <code>registral_span</code> . |
| <b>component_weighted</b>                           | <code>event_instances</code>      | Doublings, divisi unisons, and repeated noteheads <b>should</b> count (orchestral <b>mass</b> in register). <b>Primary read:</b><br><code>mean_pairwise_registral_distance</code> .                                    |

**Do not mix interpretations** across corpora: pick one profile and keep it fixed.

# 1. Static vertical aggregate

## Research question

*What is the vertical registral configuration while a given score state holds?*

Each row describes a **fixed set of sounding pitches** — no moving-window blur.

# Recommended observation mode

## event\_boundaries

- One row per maximal interval [interval\_start, interval\_end) where the active in-register pitch set is **constant**.
- window\_size is **ignored** for segmentation (still accepted by the API).
- time\_step is **unused** for indexing (still stored in exports).

# Advised parameter set

| Parameter           | UI / API value                          | Rationale                                                     |
|---------------------|-----------------------------------------|---------------------------------------------------------------|
| observation_mode    | event_boundaries                        | Exact notational vertical states                              |
| analysis_profile    | occupied_space                          | Density-independent vertical geometry                         |
| pitch_sampling_mode | (omit – implied) → unique_pitch_heights | Do not override unless you deliberately want event instances  |
| register_low        | A0                                      | Full notated range (low)                                      |
| register_high       | C8                                      | Full notated range (high)                                     |
| time_step           | 0.25                                    | Inert for segmentation; keep default for export compatibility |
| window_size         | 4.0                                     | Inert for segmentation; keep default for export compatibility |

## Primary metrics to read (per interval)

| Metric                                        | Role in static vertical reading                                                         |
|-----------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>dispersion_degree /<br/>registral_span</b> | <b>Primary</b> — canonical vertical opening (semitones); identical values               |
| <b>mean_pairwise_registral_distance</b>       | Secondary — mean pair separation (can diverge from span/degree if clustering is uneven) |
| <b>registral_centroid</b>                     | Where the vertical mass sits (absolute tessitura)                                       |
| <b>registral_std</b>                          | Tight vs loose clustering around the centroid                                           |
| <b>normalized_registral_span</b>              | Compare segments when register band is fixed                                            |
| <b>occupancy_entropy</b>                      | Optional — bin evenness; <b>not</b> dispersion (use only comparatively)                 |

## Post-processing: whole-score static summary

The tool outputs **one row per constant state**, not a single global number. For a **single aggregate** over the entire piece:

### Option A — duration-weighted mean (recommended):

```
For each interval i with duration d_i:
  weight w_i = d_i / sum(d_i)
  aggregate_span = sum(w_i * registral_span_i)  # skip NaN rows (rests)
```

Use the same weighting for `mean_pairwise_registral_distance`, `registral_centroid`, etc.

### Option B — one full-score window (single row):

Use only if you need one CSV row for the entire work:

| Parameter        | Value                                                  |
|------------------|--------------------------------------------------------|
| observation_mode | <b>fixed_window</b>                                    |
| window_size      | $\geq 2 \times \text{score\_duration}$ (quarterLength) |
| time_step        | $\geq \text{score\_duration}$                          |
| Other settings   | Same as table above                                    |

This yields essentially **one window** covering the full sounding span. Less precise than `event_boundaries` for internal registral change, but valid as a **global vertical summary**.

## Heatmap (static vertical complement)

| Parameter       | Advised value                                              |
|-----------------|------------------------------------------------------------|
| Include heatmap | <b>Yes</b>                                                 |
| Time bin size   | <b>0.25</b> qL (finer: 0.125 for short gestures)           |
| Counting mode   | <b>event_instances</b> (shows doublings / orchestral mass) |
| Color scaling   | <b>log1p_counts</b>                                        |
| Palette         | <b>registral_ember</b>                                     |

**Read:** horizontal bands = sustained register occupation; vertical extent of bright regions  $\approx$  registral footprint over time.

## UI checklist (static vertical aggregate)

- ☐ Register preset: **A0 to C8 (full notated range)**
- ☐ Temporal observation: **event\_boundaries**
- ☐ Analysis profile: **occupied\_space**
- ☐ Pitch sampling override: **(from profile)**
- ☐ Include concentration heatmap: **checked**
- ☐ Overlay mean pairwise distance: optional (secondary axis; primary curve is `dispersion_degree`)
- ☐ Plot normalized y-axis: **off** (prefer raw semitones for publication)

## CLI (static vertical aggregate)

```
python -m registral_dispersion analyze ^
--score "path/to/score.musicxml" ^
--out-dir "./out_static_vertical" ^
--prefix "static_vertical" ^
--register-low A0 ^
--register-high C8 ^
--observation-mode event_boundaries ^
--analysis-profile occupied_space ^
--plot-pairwise
```

Heatmap (separate export):

```
python -m registral_dispersion concentration-map ^
--score "path/to/score.musicxml" ^
--out "./out_static_vertical/concentration.png" ^
--register-low A0 ^
--register-high C8 ^
--time-bin-size 0.25 ^
--mode event_instances ^
--normalization log1p_counts ^
--colormap registral_ember
```

## Python API (static vertical aggregate)

```
from registral_dispersion import run_registrat_dispersion_analysis
```

```
params_static = {
    "observation_mode": "event_boundaries",
    "analysis_profile": "occupied_space",
    "register_low": "A0",
    "register_high": "C8",
    "time_step": 0.25,
    "window_size": 4.0,
}
```

```
out = run_registrat_dispersion_analysis("score.musicxml", params_static)
# Primary columns: out["results"]["registrat_span"], interval_start, interval_end
```

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## 2. Moving fragment

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### Research question

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How does registral dispersion **change over time** along the score?

Sliding windows produce a **continuous (sampled) trajectory** of vertical opening/compression.

### Recommended observation mode

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#### **fixed\_window**

- Symmetric window  $[t - w/2, t + w/2]$  at each grid point  $t$ .
- **time\_step** = grid spacing; **window\_size** = window length (both in quarterLength).

### Advised parameter set

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| Parameter           | UI / API value                   | Rationale                                                             |
|---------------------|----------------------------------|-----------------------------------------------------------------------|
| observation_mode    | <b>fixed_window</b>              | Moving temporal support                                               |
| analysis_profile    | <b>occupied_space</b>            | Registrat geometry (default)                                          |
| pitch_sampling_mode | (omit) →<br>unique_pitch_heights | —                                                                     |
| register_low        | <b>A0</b>                        | Full notated range (low)                                              |
| register_high       | <b>C8</b>                        | Full notated range (high)                                             |
| <b>time_step</b>    | <b>0.25 qL</b>                   | Standard grid: sixteenth-note at =60; good balance detail / stability |
| <b>window_size</b>  | <b>4.0 qL</b>                    | ~one bar at 4/4 =60; smooths local texture                            |

| Parameter | UI / API value | Rationale                       |
|-----------|----------------|---------------------------------|
|           |                | without hiding bar-level change |

### Alternative window sizes (same time\_step)

| Goal                                      | window_size        | time_step         | Comment                                     |
|-------------------------------------------|--------------------|-------------------|---------------------------------------------|
| <b>Fine gesture / local fragmentation</b> | <b>1.0–2.0</b> qL  | <b>0.125–0.25</b> | More reactive; noisier curves               |
| <b>Standard (advised)</b>                 | <b>4.0</b> qL      | <b>0.25</b>       | Default research balance                    |
| <b>Section / phrase level</b>             | <b>8.0–16.0</b> qL | <b>0.5–1.0</b>    | Smoother; use for long orchestral spans     |
| <b>Very long works (operas, Mahler)</b>   | <b>8.0</b> qL      | <b>0.5</b>        | Reduces row count and high-frequency jitter |

**Rule of thumb:** keep `window_size` / `time_step` between **8 and 32** (enough overlap for smooth curves without excessive redundancy).

### Primary metrics to read (moving fragment)

| Metric                                                       | Role in temporal reading                                                                                         |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>dispersion_degree</code> / <code>registral_span</code> | <b>Primary curve</b> (UI and PNG default) — smoothed vertical opening trajectory                                 |
| <code>mean_pairwise_registral_distance</code>                | Optional overlay — mean pair separation (enable when doublings matter or under <code>component_weighted</code> ) |
| <code>registral_centroid</code>                              | Tessitura drift over time (export from CSV)                                                                      |
| <code>registral_std</code>                                   | Cluster tightness evolution                                                                                      |
| <code>normalized_dispersion_degree</code>                    | Compare scores with same band                                                                                    |

## Plot options (moving fragment)

| UI option                      | Advised                                                                                        |
|--------------------------------|------------------------------------------------------------------------------------------------|
| Overlay mean pairwise distance | <b>On</b> when comparing span vs mean separation; <b>off</b> for minimal occupied-space curves |
| Show occupancy entropy         | <b>Off</b> by default – enable only for bin-evenness comparison                                |
| Plot normalized y-axis         | <b>Off</b> for single-score study; <b>On</b> for cross-score corpus                            |
| Interactive plots              | <b>On</b> (Plotly)                                                                             |
| Include heatmap                | <b>On</b> – align heatmap bin size with <code>time_step</code>                                 |

## Heatmap (moving fragment complement)

| Parameter     | Advised value                                   |
|---------------|-------------------------------------------------|
| Time bin size | <b>Same as</b> <code>time_step</code> → 0.25 qL |
| Counting mode | <code>event_instances</code>                    |
| Color scaling | <code>log1p_counts</code>                       |
| Palette       | <code>registral_ember</code>                    |

Aligning heatmap bins with dispersion grid makes visual comparison straightforward.

## UI checklist (moving fragment)

- ☐ Register preset: **A0 to C8 (full notated range)**
- ☐ Temporal observation: **fixed\_window**
- ☐ Time step / heatmap bin: **0.25**
- ☐ Window size: **4.0**
- ☐ Analysis profile: **occupied\_space**
- ☐ Include concentration heatmap: **checked**
- ☐ Overlay mean pairwise distance: optional (primary = `dispersion_degree`)
- ☐ Show occupancy entropy: **unchecked** (unless studying uniformity)

## CLI (moving fragment)

```
python -m registral_dispersion analyze ^
--score "path/to/score.musicxml" ^
--out-dir "./out_moving_fragment" ^
--prefix "moving_fragment" ^
--register-low A0 ^
--register-high C8 ^
--observation-mode fixed_window ^
--time-step 0.25 ^
--window-size 4.0 ^
--analysis-profile occupied_space ^
--plot-pairwise
```

Combined workflow (dispersion + heatmap via UI): click **Run analysis** once with the settings above.

## Python API (moving fragment)

```
from registral_dispersion import run_registrat_dispersion_analysis

params_moving = {
    "observation_mode": "fixed_window",
    "analysis_profile": "occupied_space",
    "register_low": "A0",
    "register_high": "C8",
    "time_step": 0.25,
    "window_size": 4.0,
}

out = run_registrat_dispersion_analysis("score.musicxml", params_moving)
# Primary curve: out["results"]["dispersion_degree"] vs out["results"]["t"]
```

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## Quick comparison

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|                                      | Static vertical aggregate                                        | Moving fragment                                                      |
|--------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Observation</b>                   | event_boundaries                                                 | fixed_window                                                         |
| <b>Time logic</b>                    | One row per <b>constant pitch set</b>                            | One row per <b>sliding window</b>                                    |
| <b>Primary metric</b>                | dispersion_degree /<br>registral_span                            | dispersion_degree (curve);<br>pairwise optional overlay              |
| <b>Profile</b>                       | occupied_space                                                   | occupied_space                                                       |
| <b>Register</b>                      | A0–C8                                                            | A0–C8                                                                |
| <b>time_step</b>                     | 0.25 (inert)                                                     | <b>0.25</b> (active grid)                                            |
| <b>window_size</b>                   | 4.0 (inert)                                                      | <b>4.0</b> (active)                                                  |
| <b>Heatmap <math>\Delta t</math></b> | 0.25                                                             | <b>0.25</b> (= time_step)                                            |
| <b>Best for</b>                      | Segment taxonomy, vertical<br>states, orchestration<br>snapshots | Trajectories, climaxes, registral<br>expansion/contraction over time |

### 3. One-number global summary

#### Research question

*What is a single whole-score registral dispersion value for this symbolic score?*

#### Recommended path

summarize\_registral\_dispersion (API) or python -m registral\_dispersion  
summarize (CLI).

| Parameter                    | Advised value                               |
|------------------------------|---------------------------------------------|
| observation_mode             | event_boundaries                            |
| analysis_profile             | occupied_space                              |
| register_low / register_high | A0 / C8                                     |
| tie_policy                   | as_imported (or merge_ties for tied scores) |

Primary: `duration_weighted_registral_span`. Secondary:  
`duration_weighted_mean_pairwise_registral_distance`.

```
python -m registral_dispersion summarize --score "path/to/score.musicxml" --c
```

```
from registral_dispersion import summarize_registral_dispersion
out = summarize_registral_dispersion("score.musicxml")
print(out["primary_metric"], out["primary_value"], out["warnings"])
```

Do not use `fixed_window` for a duration-weighted one-number unless you accept a sampled trajectory summary (warning emitted).

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## When to switch to `component_weighted`

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Use `analysis_profile: component_weighted` (implies `event_instances`) if the research question explicitly includes:

- double octaves and massed unisons;
- divisi doubling at the same pitch;
- orchestrational **weight** rather than pure registral geometry.

Then treat `mean_pairwise_registral_distance` as primary in both setups.

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## Reproducibility checklist (publication)

- 
1. Record **package\_version** from JSON export.
  2. Archive **exact parameter dict** (this document + your score path).
  3. State **register band** and **profile** in the paper methods section.
  4. Distinguish **dispersion metrics** from **occupancy\_entropy** and from **concentration-map notational occupancy** (not acoustic density). See [METRIC SEMANTICS.md](#).
  5. Record **tie\_policy**, **global\_summary**, and any **warnings** from JSON export.
  6. Run **benchmarks/scripts/compare\_frozen\_outputs.py** after intentional code changes (synthetic fixtures only).
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*Document version: 2026-06-03 — matches Registrat\_Dispersion / registrat-dispersion 0.3.0 (plotting: primary dispersion\_degree; JSON schema 1.8).*

